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In one species of squirrel, *Sciurus carolinensis*, fur colour is controlled by one gene, with two codominant alleles. C^G represents the allele for grey fur colour, and C^B represents the allele for black fur colour.

Table 1 shows the three possible phenotypes.

Table 1

Genotype	Phenotype
$C^G C^G$	Grey fur
$C^G C^B$	Brown-black fur
$C^B C^B$	Black fur

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In a population of 34 *S. carolinensis*, 2 had black fur.

Use the Hardy–Weinberg equation to estimate how many squirrels in this population had brown-black fur. Show your working.

[2 marks]

Answer _____

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The actual number of squirrels in this population that had brown-black fur was 16.

Use all of the information to calculate the **actual** frequency of the C^G allele.

Do **not** use the Hardy–Weinberg equation in your calculation.

Give your answer to 2 decimal places.

[1 mark]

Answer _____

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S. carolinensis were first introduced to the UK from North America in the 1870s. They are now widely distributed across the UK.

S. carolinensis from both North America and the UK show exactly the same genotypic and phenotypic variation. An identical mutation causing black fur has also been found in several other species closely related to *S. carolinensis*.

Use this information to deduce which **one** of the following conclusions is most likely true.

Tick (✓) **one** box.

[1 mark]

- A** The mutation that caused black fur happened after *S. carolinensis* was introduced to the UK from North America.
- B** The mutation that caused black fur happened in a common ancestor of *S. carolinensis* and other closely related species.
- C** The mutation that caused black fur happened independently in *S. carolinensis* and all other closely related species.
- D** The phenotypic variation shown in *S. carolinensis* and other closely related species is caused by genetic drift.

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Question 1 continues on the next page

The mutation that caused the C^B allele was due to a 24 base-pair deletion from the C^G allele.

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The protein coded for by the C^B allele is 306 amino acids long.

Calculate the percentage reduction in size of the protein coded for by the C^B allele compared with the protein coded for by the C^G allele.

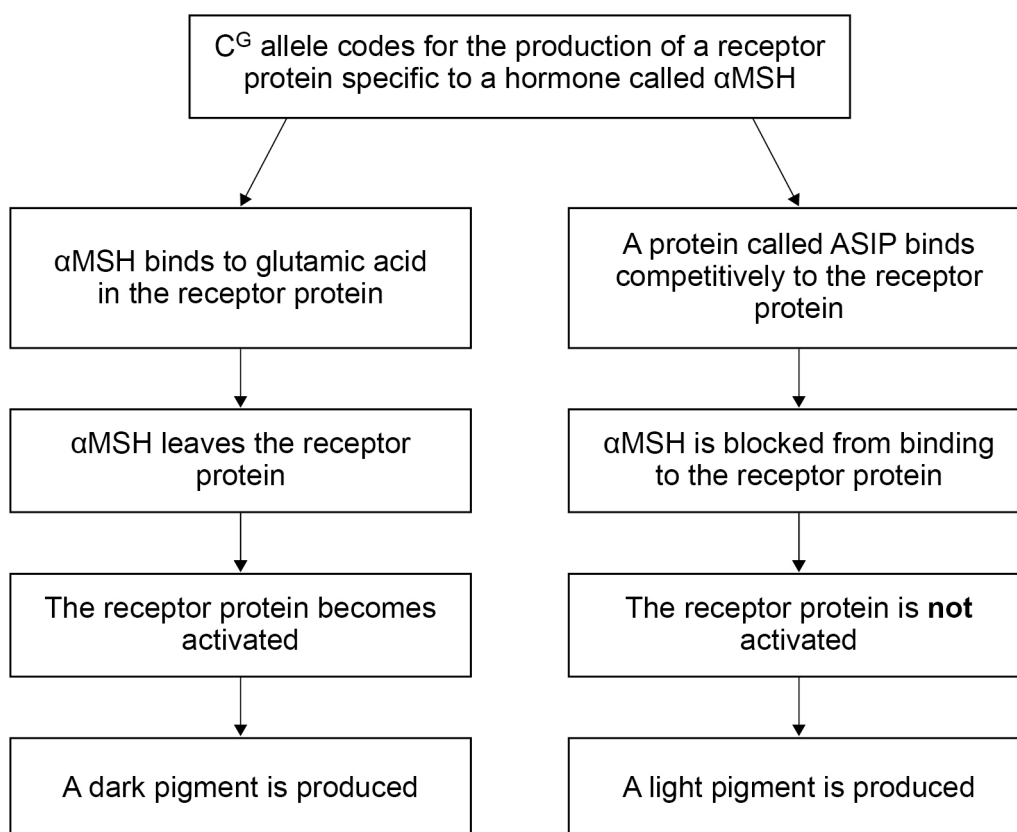
Give your answer to 3 significant figures and show your working.

[2 marks]

Answer _____

In *S. carolinensis*, fur colour depends on the distribution and relative amounts of light pigments and dark pigments in the hairs of the fur. **Figure 1** shows how the protein produced from the C^G allele can result in the production of a light pigment or a dark pigment.

Figure 1



The deletion mutation in the C^B allele results in the production of a receptor protein that does not have glutamic acid. The lack of glutamic acid in the receptor protein has the same effect as α MSH leaving the receptor protein.

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Use **Figure 1** and this information to suggest why *S. carolinensis* with the genotype $C^B C^B$ have black fur rather than grey fur.

[3 marks]

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Turn over for the next question

